

Homework 1

Release Date: 1404/7/30

Deadline: 1404/8/8

(*) starred problems are optional and have a bonus mark!

1 Fourier Series (12+3pts)

Determine the Fourier series representation of the following periodic signals.

1.
$$x(t) = \frac{1}{\sqrt{2\pi\sigma^2}} \sum_{n=-\infty}^{\infty} e^{-\frac{(t-\mu-nT)^2}{2\sigma^2}}.$$

2.
$$x(t) = |\cos(2\pi f_0 t)|.$$

3.
$$x(t) = \sum_{n=-\infty}^{\infty} e^{t-n} [u(t-n) - u(t-(n+1))].$$

4.
$$x(t) = \cos(2\pi t) + \cos(4\pi t).$$

5. (*)
$$x(t) = \sum_{n=-\infty}^{\infty} \Lambda(t-n) u(t-n).$$

2 Fourier Transform (12pts)

Compute the continuous-time Fourier transforms of the following signals for $t \in \mathbb{R}$, $f_0 > 0$, $a > 0$, and $\alpha > 0$.

1.
$$x(t) = \frac{t}{(a^2 + t^2)^2}.$$

2.
$$x(t) = 4 \operatorname{rect}\left(\frac{t}{4}\right) \cos(2\pi f_0 t).$$

3.
$$x(t) = t \operatorname{sinc}(t).$$

4.
$$x(t) = t e^{-\alpha|t|} \cos(\beta t).$$

3 Type of Signals (15pts)

Classify each of the following signals as an energy signal, a power signal, or neither. For those signals that are energy or power type, compute their corresponding energy or power values.

1.
$$x(t) = e^{-\alpha|t|} \cos(\beta t), \quad (\alpha > 0).$$

2. $x(t) = e^{-t} \cos t$.

3. $x(t) = \text{sgn}(t)$.

4. $x(t) = \begin{cases} K t^{-\frac{1}{4}}, & t > 0, \\ 0, & t \leq 0. \end{cases}$

5. $x(t) = \frac{1}{\pi t}$.

4 Fourier Transform and Real Integrals (16pts)

Using the properties of the Fourier transform, evaluate the following integrals ($\alpha > 0$).

1. $I = \int_{-\infty}^{+\infty} \text{sinc}^5(t) dt$

2. $I = \int_0^{+\infty} e^{-\alpha t} \text{sinc}(t) dt$

3. $I = \int_0^{+\infty} e^{-\alpha t} \cos(\beta t) dt$

4. $I = \int_{-\infty}^{+\infty} \frac{\sin(t) - t \cos(t)}{t^3} dt$

5 Energy Signals (15pts)

Let $x(t)$ and $y(t)$ be two energy signals with corresponding energies E_x and E_y .

1. If $x(t)$ and $y(t)$ are orthogonal, show that the energy of the signal $x(t) + y(t)$ is equal to the sum of the individual energies $E_x + E_y$.
2. If $x(t)$ and $y(t)$ are orthogonal, find the energies of the signals $c_1 x(t) + c_2 y(t)$ and $c_1 x(t) - c_2 y(t)$.
3. For two energy signals $x(t)$ and $y(t)$, their cross-energy is defined as

$$E_{xy} = \int_{-\infty}^{+\infty} x(t) y^*(t) dt.$$

If $z(t) = x(t) \pm y(t)$, show that

$$E_z = E_x + E_y \pm (E_{xy} + E_{yx}).$$

6 Poisson's Sum Formula (20pts)

1. By computing the Fourier series coefficients for the periodic signal

$$s(t) = \sum_{n=-\infty}^{+\infty} \delta(t - nT_s),$$

show that

$$\sum_{n=-\infty}^{+\infty} \delta(t - nT_s) = \frac{1}{T_s} \sum_{n=-\infty}^{+\infty} e^{jn \frac{2\pi t}{T_s}}.$$

2. Using the result of part (1), prove that for any signal $x(t)$ and any sampling period T_s , the following identity holds

$$\sum_{n=-\infty}^{+\infty} x(t - nT_s) = \frac{1}{T_s} \sum_{n=-\infty}^{+\infty} X\left(\frac{n}{T_s}\right) e^{jn\frac{2\pi t}{T_s}}.$$

3. Conclude the following relation, known as the Poisson's Sum Formula

$$\sum_{n=-\infty}^{+\infty} x(nT_s) = \frac{1}{T_s} \sum_{n=-\infty}^{+\infty} X\left(\frac{n}{T_s}\right).$$

4. Using Poisson's sum formula, show that:

$$\sum_{k=-\infty}^{+\infty} \frac{2\alpha}{\alpha^2 + 4\pi^2 k^2} = \coth\left(\frac{\alpha}{2}\right), \quad (\alpha > 0).$$

7 Parseval's Theorem (10+7pts)

Parseval's energy theorem states that the integral of the squared magnitude of a signal (i.e., the signal's total energy) in the time domain is equal to the integral of the squared magnitude of its Fourier transform in the frequency domain, i.e.,

$$E = \int_{-\infty}^{+\infty} |x(t)|^2 dt = \int_{-\infty}^{+\infty} |X(f)|^2 df.$$

Also to compute the **average energy** of a periodic signal, parseval's theorem can be written as

$$\frac{1}{T} \int_T |x(t)|^2 dt = \sum_{k=-\infty}^{+\infty} |a_k|^2,$$

where a_k represents the Fourier series coefficients of the periodic signal $x(t)$ with period T .

1. Using the properties of the Fourier transform and Parseval's theorem, evaluate the following integral

$$\int_0^{+\infty} \frac{1}{(1+x^2)^2} dx.$$

2. (*) Suppose that the input signal to an LTI system is given by

$$x_1(t) = \sum_{k=-\infty}^{+\infty} \alpha^{|k|} e^{jk\frac{\pi}{2}t},$$

where $0 < \alpha < 1$ and the frequency response of the system is given by

$$H(f) = \begin{cases} 1, & |f| < W, \\ 0, & \text{otherwise.} \end{cases}$$

Determine the minimum value of W such that the average energy of the output signal is at least 90% of the average energy of the input signal.

Submission

Please submit your answers in a .pdf file named **CS_HW1_studentID.pdf**, replacing **studentID** with your actual student ID.